Quillen-Suslin

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May 2026

Section 3 is based closely off Chapter XXI of Lang's *Algebra* [Lan02]; a main difference is that we are more liberal in our use of Noetherian hypotheses in Section 3.1.

1 Motivation

Definition 1.0.1. Recall that a *(complex) topological (resp. smooth, holomorphic, algebraic) vector bundle of rank* r over a topological space (resp. smooth manifold, complex manifold, complex algebraic variety) X is a total space E and structure map $\pi : E \to X$ such that X admits a cover $\mathcal{U} := \{U_i\}$ for which

1. there are homeomorphisms (resp. diffeomorphisms, biholomorphisms, isomorphisms) $\varphi_i : \pi^{-1}(U_i) \xrightarrow{\sim} \mathbb{C}^r \times U_i$ making the triangle

$$\begin{tikzcd}
\pi^{-1}(U_i) \arrow[rr, "\varphi_i"] \arrow[dr, "\pi"'] & & \mathbb{C}^r \times U_i \arrow[dl] \\
& U_i &
\end{tikzcd}$$

   commute,

2. for each pair (i, j), the composition $\varphi_j \circ \varphi_i^{-1} : \mathbb{C}^r \times (U_i \cap U_j)$ is a linear isomorphism on each fiber,

3. and these isomorphisms vary nicely, i.e. the map $T_{ij} : U_i \cap U_j \to \mathrm{GL}_r(\mathbb{C})$ sending a point p to the isomorphism $(\varphi_j \circ \varphi_i^{-1})|_{\mathbb{C}^r \times \{p\}}$ is continuous (resp. smooth, holomorphic, algebraic).

Example 1.0.2. Given any space X and any $r \in \mathbb{Z}_{\geq 0}$, we may construct the product $X \times \mathbb{C}^r$ and consider the projection map $\pi : X \times \mathbb{C}^r \to X$. This is a rank r vector bundle over X, called the *trivial bundle*.

Example 1.0.3. Below is a depiction of the Möbius bundle, a real rank 1 vector bundle over the circle. This is about the only non-trivial example that one can really draw.

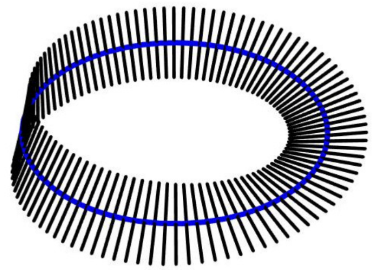

Figure 1: A Möbius bundle. The black line segments are the fibers; one should imagine them continuing off to infinity. Image taken from [Lyn].

Definition 1.0.4 (Pullback)**.** Let $f : Y \to X$, and let $\pi : E \to X$ be a vector bundle of rank r. Then, we may obtain a vector bundle over Y, denoted $f^*(E)$ and called the *pullback of E along f*, by "declaring the fiber over each point p of Y to be the fiber of π over $f(p)$." Precisely, $f^*(E) := Y \times_X E$, and the structure map $\tilde{\pi} : f^*(E) \to Y$ is the map induced by the fiber product. Concretely, $f^*(E) = \{(q, v) \in Y \times E : f(q) = \pi(v)\}$, and $\tilde{\pi}(q, v) = q$.

Theorem 1.0.5. Let $\pi : E \to X$ be a topological (resp. smooth) vector bundle on a paracompact space, and let $f_0, f_1 : Y \to X$ be homotopic maps. Then, the pullbacks $f_0^*(E), f_1^*(E)$ are isomorphic.

Proof. A full proof can be found in [Hat17, pp. 20–21]. The idea is to pull back along the whole homotopy F, and then to build painstakingly build isomorphisms $f_{t_i}^*(E) \xrightarrow{\sim} f_{t_{i+1}}^*(E)$ using partitions of unity. $\square$

Corollary 1.0.6. Topological and smooth vector bundles on contractible paracompact spaces are trivial.

Seeing the nice result for topological and smooth vector bundles, one may next ask about the next most flexible setting, holomorphic bundles. The holomorphic setting is too rigid for a partition of unity argument to go through, and, indeed, in this setting things break down.

Theorem 1.0.7. The contractible open submanifold

$$M := \{(z, w) \in \mathbb{C}^2 : \max(|z|, |w|) < 2, \quad \min(|z|, |w|) < 1\}$$

of $\mathbb{C}^2$ admits a non-constant holomorphic line bundle.

Proof. See [Spe]. $\square$

The above example, being ball-like, has a very non-algebraic flavor. Nonetheless, there are also contractible smooth complex algebraic varieties which admit non-trivial vector bundles; see [AD08].

So, homotopy-invariance of classification of vector bundles breaks down in more rigid categories. Interestingly, in the holomorphic setting we may save it by restricting our class of manifolds.

Theorem 1.0.8 (Oka-Grauert Principle)**.** The forgetful map

$$\{\text{isom. classes of holomorphic V.B.s}\} \longrightarrow \{\text{isom. classes of topological V.B.s}\}$$

is a bijection for closed complex submanifolds of $\mathbb{C}^n$ (known as *Stein manifolds*) [For11, p. 190].

Corollary 1.0.9. There are no non-trivial holomorphic vector bundles on contractible Stein manifolds.

Stein manifolds are sort of the complex analytic analogues of affine varieties. So, seeing the Oka-Gauert principle come out in the late 1950's, it would be natural for algebraic geometers of the time to ask about vector bundles on contractible affine varieties, and to start with the canonical example, $\mathbb{A}^n$. A proof would allude them for about twenty years.

Theorem 1.0.10 (Quillen-Suslin, 1976)**.** There are no non-trivial vector bundles on $\mathbb{A}^n_k$.

2 Vector bundles and projective modules

Let $X = \operatorname{Spec} R$ be an affine variety over a field k. Let $\pi : E \to X$ be a rank r vector bundle over X, and let $\Gamma(E, X)$ denote the space of algebraic sections $\sigma : X \to E$ of π. Using the vector space structure on the fibers of π, we may add sections pointwise. Furthermore, we may scale sections pointwise according to functions $f \in R$. Thus, $\Gamma(E, X)$ is an R-module.

We now describe the possible R-modules arising in this way.

Theorem 2.0.1. The modules of sections of rank r vector bundles on X are exactly the locally free modules of rank r over R.

Proof. $\underline{\implies}$ We first show the module of sections of a vector bundle is locally free. Fix $\pi : E \to X$. Since vector bundles are locally trivial, there are $U_1, ..., U_n$ covering X such that $E_i := E|_{U_i}$ is trivial. By refining our cover if necessary, we may assume the U_i are the complements $D(f_i)$ of the vanishings of functions.

On one hand, sections $\sigma : U_i \to E_i$ are the same as rational global sections vanishing outside U_i, so $\Gamma(E_i, U_i) = \{f_i^n\}^{-1}\Gamma(E, U)$. On the other hand, sections $\sigma : U_i \to U_i \times k^r$ of a product are the same data as maps $U_i \to k^r$, which are the same data as r-tuples of functions $U_i \to k$. That is, $\Gamma(E_i, U_i) \cong R[f^{-1}]^r$ (as $R[f^{-1}]$-modules). Putting these two perspectives together, we glean that $\Gamma(E, X)$ is a locally free R-module of rank r.

$\underline{\impliedby}$ Conversely, let M be a locally free R-module of rank r. Let

$$\operatorname{Sym}^\bullet_R M^\vee := R \oplus M^\vee \oplus \operatorname{Sym}^2_R(M^\vee) \oplus ...$$

denote the symmetric tensor algebra of $M^\vee$, and let $E := \operatorname{Spec}(\operatorname{Sym}^\bullet M^\vee)$. Define $\pi : E \to X$ to be the map of spectra corresponding to the inclusion of R as the degree 0 piece of piece of $\operatorname{Sym}^\bullet_R(M^\vee)$. We claim $\pi : E \to X$ is a rank r vector bundle with module of sections M.

We start with the "vector bundle of rank r" part. As we've defined them, vector bundles are free on an open cover, whereas locally free modules are just free at each *stalk*. We now show the somewhat standard fact that for finite freeness, these two notions of locality agree. Free on opens of course implies free on stalks, so we really just show the other direction.

Lemma 2.0.2. Let M be an R module, and let p be a prime of R such that M_p is a finitely generated free R_p module. Then, there is some $f \in R\backslash p$ for which $\{f^n\}^{-1}M$ is a free $R[f^{-1}]$-module.

Proof. By hypothesis, M_p is free of rank r, say with basis $m_1, ..., m_r$. We have a map

$$\begin{aligned} \varphi : \qquad R^r &\longrightarrow \qquad M \\ (a_1, ..., a_r) &\longmapsto \sum a_i m_i. \end{aligned}$$

The modules $\ker\varphi$ and $\operatorname{coker}\varphi$ are a sub and a quotient respectively of finitely generated modules over a Noetherian ring, so each is finitely generated. Furthermore, since the localized map $R_p^r \to M_p$ is an isomorphism, $\ker\varphi$ and $\operatorname{coker}\varphi$ are not supported at p.

Finitely generated modules over Noetherian rings have closed support, so $\ker\varphi$ and $\operatorname{coker}\varphi$ are supported at $V(I)$ and $V(J)$ respectively, for $I, J \not\subseteq p$. Pick finite generating sets for I and J, and let f be the product of all those generators. Then, $\varphi_f : R[1/f]^r \to M[1/f]$ has kernel and cokernel supported nowhere, making φ_f an isomorphism, as desired. ∎

Having established that E is a rank r vector bundle on X, we now show that $\Gamma(E, X) \cong M$. A section is map $\sigma : X \to E$ such that $\pi \circ \sigma = \mathrm{id}_X$. On the level of rings, this is a diagram

$$\begin{tikzcd} R \arrow[r, "\pi^\#"] \arrow[rr, bend right, "\mathrm{id}_R"'] & \operatorname{Sym}^\bullet_R(M^\vee) \arrow[r, "\sigma^\#"] & R. \end{tikzcd}$$

The condition that the diagram commutes is exactly the requirement that $\sigma^\#$ be a map of R-algebras. The middle ring $\operatorname{Sym}^\bullet_R(M^\vee)$ is generated as an R-algebra in degree 1, so $\sigma^\#$ is determined by what it does to M. Conversely, any R-module homomorphism $M \to R$ extends to a map $\operatorname{Sym}^\bullet_R(M^\vee) \to R$ of algebras. Thus, $\Gamma(E, X) \cong \operatorname{Hom}(M^\vee, R) =: M^{\vee\vee}$.

We wanted $\Gamma(E, X) \cong M$, so what is left is to show that $M \cong M^{\vee\vee}$. One always has a map to the double dual, as follows.

$$M \longrightarrow M^{\vee\vee}, \qquad m \longmapsto (\varphi \mapsto \varphi(m)).$$

For general R-modules, this need not be an isomorphism. However, the canonical map to the double dual *is* an isomorphism for finite rank free modules. Since M is locally free of finite rank, and the property "is an isomorphism" can be checked locally, the canonical map $M \to M^{\vee\vee}$ is an isomorphism. $\square$

Recall from your course on commutative algebra that projective modules are those for which $\mathrm{Hom}(P, -)$ is exact. Recall also that finitely presented modules are projective if and only if they are locally free. Since $k[x_1, ..., x_n]$ is Noetherian, finite presentation is the same as finite generation.

Corollary 2.0.3. Isomorphism classes of vector bundles on $\mathbb{A}^n_k$ are in bijective correspondence with isomorphism classes of finitely generated projective modules over $k[x_1, ..., x_n]$.

3 A proof of the Quillen-Suslin theorem

In accordance with the previous section, we reformulate the problem of triviality of vector bundles on $\mathbb{A}^n_k$ as that of freeness of finitely generated projective modules on $k[x_1, ..., x_n]$. The proof breaks down into three main steps.

Theorem 3.0.1. Finitely generated projective modules over $k[x_1, ..., x_n]$ are *stably free.*

Theorem 3.0.2. The ring $k[x_1, ..., x_n]$ has the *unimodular column extension property.*

Theorem 3.0.3. Stably free modules over rings with the universal column extension property are free.

Included in these theorems are two yet-to-be-defined terms; we delay both definitions to their respective sections. Also, while we just listed the theorems in the nicest order for a syllogism, we prove them in the order that makes each feel most motivated, tackling 3.0.1, then 3.0.3, then 3.0.2.

3.1 Projective vs. stably free

NOTE: Throughout this section, all rings are Noetherian, and all modules are finitely generated. Also, we recommend skipping the proofs of the lemmas on a first read; they will be better appreciated once one knows what they aim to build towards.

Definition 3.1.1 (Stably free)**.** A module M is *stably free* is there is a finite rank free module F for which $M \oplus F$ is finite free, meaning free and finitely generated.

The existence of stably free modules that are not free is initially surprising to many. Some intuition can be gleaned by comparing to stably trivial vector bundles. A vector bundle E is *stably trivial* if $E \oplus F$ is trivial for some trivial bundle F.

The famous hairy ball theorem gives, as a quick corollary, that the tangent bundle TS^2 to the sphere is non-trivial. The normal bundle, on the other hand, can be shown rather quickly to be trivial. Their direct sum is the restriction of the tangent bundle of $\mathbb{R}^3$ to S^2, so the direct sum is trivial of rank 3. Thus, TS^2 is a non-trivial stably trivial vector bundle. As it so happens, this topological example can be translated into an algebraic example.

Example 3.1.2. Let $R = \mathbb{R}[x, y, z]/(x^2 + y^2 + z^2 - 1)$. Define the submodule $T := \{(f, g, h) \in R^3 : xf + yg + zh = 0\}$. Then, T is stably free but not free, with $T \oplus R \cong R^3$ but $T \ncong R$.

Proof. See [Con] for details. $\square$

Stably free modules are by definition direct summands of free modules, hence projective. The goal of this subsection is to show that over $k[x_1, ..., x_n]$, the converse holds. The gameplan is to prove a statement about all modules that specializes to what we want for projective modules. We first introduce some theory regarding finite free resolutions (see Definition 3.1.4), culminating in Theorem 3.1.16. We then prove the main result by induction on the number of variables. Theorem 3.1.16 is crucial in this step, allowing us to make all kinds of reductions, for example to the case of R a domain and M a prime ideal of $R[x]$ whose intersection with R is trivial.

Remark 3.1.3. Theorem 3.0.1 was proven by Serre in the 50's, long before the full Quillen-Suslin theorem. One thing that makes the second half harder is that the statement

"Finitely generated projective modules on R are free

$$\Downarrow$$

Finitely generated projective modules on $R[x]$ are free"

is simply not true – the localization of $k[x, y]/(y^2 - x^3)$ at (x, y) is a counterexample; see [Wei13, p. 23]. Thus, an induction on number of variables could not work.

Definition 3.1.4 (Finite free resolution)**.** A *finite free resolution of length* n of an R-module M is an exact sequence

$$0 \to F_n \to F_{n-1} \to ... \to F_1 \to F_0 \to M \to 0,$$

where each F_i is finite free.

Lemma 3.1.5. Let P be projective. Then, P admits a finite free resolution if and only if P is stably free.

Proof. Suppose P is stably free. Then, $P \oplus F_1 = F_0$ for some F_0, F_1 finite free. This gives an exact sequence

$$0 \to F_1 \to F_0 \to P \to 0,$$

establishing that P admits a finite free resolution. For the other direction, we proceed by induction on the length of the finite free resolution. A finite free resolution of length 0 is an isomorphism with a free module. For $n > 0$, assume all projective modules admitting finite free resolutions of length up to $n-1$ are stably free, and suppose P admits a finite free resolution

$$0 \to F_n \to F_{n-1} \to \ldots \to F_1 \to F_0 \to P \to 0.$$

By exactness, $\ker(F_0 \to P) = \operatorname{im}(F_1 \to F_0)$. Call this single module Q. We get a decomposition of the above exact sequence as

$$\begin{aligned} &0 \to F_n \to F_{n-1} \to \ldots \to F_1 \to Q \to 0 \\ \text{and} \qquad &0 \to Q \to F_0 \to P \to 0. \end{aligned}$$

Since P is projective, the second exact sequence splits, giving $P \oplus Q = F_0$. In particular, Q is projective. The first exact sequence is a finite free resolution of length $n-1$ for Q, so by the inductive hypothesis Q is stably free. Thus, write $Q \oplus E = E'$ for finite free E, E'. Direct sum both sides with E, giving

$$P \oplus E' = P \oplus Q \oplus E = F_0 \oplus E.$$

The righthand side is a sum of frees, hence free, so we conclude that P is stably free. □

Definition 3.1.6. We define a *stably free resolution of length* n analogously to a finite free resolution; relax the finite free condition on the resolving modules to stably free.

Lemma 3.1.7. Let M be an R-module, and $n \geq 1$. Then, M admits a finite free resolution of length n if and only if M admits a stably free resolution of length n.

Proof. A finite free resolution of length n constitutes a stably free resolution of length n. Towards the other direction, let

$$0 \to E_n \to E_{n-1} \to \ldots \to E_1 \to E_0 \to M \to 0$$

be a stably free resolution of M. Choose $\{F_i\}$ such that $E_i \oplus F_i = F_i'$ is free. We are going to throw in the F_i's in pairs to make everything free while keeping everything exact. Precisely, if $n = 1$, then

$$0 \to E_1 \oplus F_1 \oplus F_0 \to E_0 \oplus F_1 \oplus F_0 \to M \to 0$$

is a finite free resolution of M of length 1. Otherwise, let

$$F_i'' = \begin{cases} E_0 & \text{if } i = 0 \\ E_{n-1} \oplus F_n \oplus F_{n-1} \oplus F_{n-2} & \text{if } i = n-1 \\ E_i \oplus F_i \oplus F_{i-1} & \text{otherwise,} \end{cases}$$

and then

$$0 \to F_n'' \to F_{n-1}'' \to \ldots \to F_1'' \to F_0'' \to M \to 0$$

is a finite free resolution of M of length n. □

The following lemma is used in the proof of the lemma after it, and nowhere else.

Lemma 3.1.8 (Schanuel's Lemma)**.** Let

$$0 \to K \to P \to M \to 0$$

$$0 \to K' \to P' \to M \to 0$$

be two short exact sequences arising from surjections of projective modules onto M. Then, $K \oplus P' \cong K' \oplus P$.

Proof. Since P is projective and $P' \to M$ is surjective, we get the following commutative diagram.

$$\begin{tikzcd}
0 \arrow[r] & K \arrow[r, "i"] & P \arrow[r, "q"] \arrow[d, "\varphi"] & M \arrow[r] \arrow[d, leftrightarrow, "\mathrm{id}_M"] & 0 \\
0 \arrow[r] & K' \arrow[r, "j"'] & P' \arrow[r, "p"'] & M \arrow[r] & 0
\end{tikzcd}$$

By exactness of the top row, $i(K) = \ker q$. By commutativity of the right square, $(\varphi \circ i)(K) \subseteq \ker p$. By exactness of the bottom row, $\ker p = j(K')$. Since j in injective, we canonically identify $j(K')$ and K', giving a unique map $\psi : K \to K'$ making the following whole diagram commute.

$$\begin{tikzcd}
0 \arrow[r] & K \arrow[r, "i"] \arrow[d, "\psi"] & P \arrow[r, "q"] \arrow[d, "\varphi"] & M \arrow[r] \arrow[d, leftrightarrow, "\mathrm{id}_M"] & 0 \\
0 \arrow[r] & K' \arrow[r, "j"'] & P' \arrow[r, "\rho"'] & M \arrow[r] & 0
\end{tikzcd}$$

Now consider the sequence of maps

$$0 \to K \xrightarrow{\left[\begin{smallmatrix} i \\ \psi \end{smallmatrix}\right]} P \oplus K' \xrightarrow{[\varphi \ -j]} P' \to 0.$$

Once we show this sequence is exact, projectivity of P' will complete the proof. Towards exactness, we first observe that one factor of $K \to P \oplus K'$ is injective, and one factor of $P \oplus K' \to P'$ is surjective, so the sequence is exact at K and P'. We next show the composition $K \to P'$ is zero. This is just computation:

$$\begin{bmatrix} \varphi & -j \end{bmatrix} \begin{bmatrix} i \\ \psi \end{bmatrix} = \begin{bmatrix} i \circ \varphi - \psi \circ j \end{bmatrix} = [0],$$

with the last equality being commutativity of the left square. Finally, we show $\ker(P \oplus K' \to P') \subseteq \mathrm{im}(K \to P \oplus K)$. Let $(p, k) \in \ker(P \oplus K' \to P')$. Then, $\varphi(p) = j(k)$. In particular, $\varphi(p) \in \mathrm{im}\, j = \ker \rho$. By commutativity of the right square, $p \in \ker q = \mathrm{im}\, i$, so we may write $p = i(\ell)$ for some $\ell \in K$. By commutativity of the left square, $j(\psi(\ell)) = j(k)$. By injectivity of j, we conclude $\psi(\ell) = k$. Bringing everything together, we get that $(k, p) = (i, \psi)(\ell) \in \mathrm{im}(K \to P \oplus K')$, as desired.

$\square$

Definition 3.1.9. We say two modules M_1, M_2 are *stably isomorphic* if there are finite free modules F_1, F_2 for which $M_1 \oplus F_1 \cong M_2 \oplus F_2$.

Remark 3.1.10. Notice that F_1, F_2 need not be equal rank. In particular, all stably free modules are stably isomorphic to one another.

Lemma 3.1.11. Let M_1, M_2 be stably isomorphic and E_1, E_2 stably free, fitting in short exact sequences

$$0 \to K_1 \to E_1 \to M_1 \to 0$$

$$0 \to K_2 \to E_2 \to M_2 \to 0.$$

Then, K_1 and K_2 are stably isomorphic.

Proof. With Schanuel's Lemma in hand, we essentially just unfold definitions and follow our noses. Stable isomorphism of M_1 and M_2 means there are finite free F_1, F_2 for which $M_1 \oplus F_1 \cong M_2 \oplus F_2$. Direct sum each sequence with $0 \to 0 \to F_i \to F_i \to 0$ to get

$$0 \to K_1 \to E_1 \oplus F_1 \to M_1 \oplus F_1 \to 0$$

$$0 \to K_2 \to E_2 \oplus F_2 \to M_2 \oplus F_2 \to 0.$$

The E_i are stably free, hence projective, so the hypotheses of Schanuel's Lemma are satisfied; the conclusion is that

$$K_1 \oplus E_1 \oplus F_1 \cong K_2 \oplus E_2 \oplus F_2.$$

Take free modules F_1', F_2' trivializing E_1 and E_2. Then, the equality

$$K_1 \oplus (E_1 \oplus F_1 \oplus F_1' \oplus F_2') \cong K_2 \oplus (E_2 \oplus F_2 \oplus F_1' \oplus F_2')$$

of enormous direct sums furnishes the desired stable isomorphism between K_1 and K_2. $\square$

Definition 3.1.12 (Stably free dimension)**.** The *stably free dimension* of an R-module M is the minimal length of a stably free resolution of M. When M is not itself stably free, by Lemma 3.1.7, the stably free dimension is also the minimal length of a finite free resolution of M.

Lemma 3.1.13. Let M have stably free dimension $n > m$, and let

$$E_m \to E_{m-1} \to ... \to E_1 \to E_0 \to M \to 0$$

be an exact sequence, with each E_i stably free. Then, there exists a stably free resolution

$$0 \to E_n \to E_{n-1} \to ... \to E_1 \to E_0 \to M \to 0$$

of length n extending the original exact sequence.

Proof. Let
$$0 \to F_n \to F_{n-1} \to ... \to F_1 \to F_0 \to M \to 0$$
be a finite free resolution for M, and let

$$... \to E_n \to E_{n-1} \to ... \to E_1 \to E_0 \to M \to 0$$

be any extension of the original sequence to a(n *a priori* infinite) resolution of M by stably free modules. Let both E_{-1} and F_{-1} denote M, and define $E_{-2} = F_{-2} = 0$. For $i \geq -1$, let $K_i = \ker(E_i \to E_{i-1})$ and $L_i = \ker(F_i \to F_{i-1})$. We will now show K_i and L_i are stably isomorphic.

We proceed by induction. For $i = -1$, we just have $K_{-1} \cong M \cong L_{-1}$. For the inductive step, notice that by exactness we have short exact sequences

$$0 \to K_i \to E_i \to K_{i-1} \to 0$$

$$0 \to L_i \to F_i \to L_{i-1} \to 0.$$

By Lemma 3.1.11, we get K_i stably isomorphic to L_i. Of interest is the case $i = n$. Here, $L_n = 0$, so K_n is stably trivial. Thus, we may cut off the resolution $E_\bullet$, taking
$$0 \to L_n \to E_{n-1} \to ... \to E_1 \to E_0 \to M \to 0.$$

□

Lemma 3.1.14. Let $0 \to N \to E \to M \to 0$ be exact, with E stably trivial. Suppose M has stably trivial dimension $n \geq 1$. Then, N has stably trivial dimension $n - 1$.

Proof. For one bound, observe that any stably free resolution

$$0 \to E_{n-1} \to ... \to E_1 \to E_0 \to N \to 0$$

can be glued to the starting short exact sequence along N to yield a stably free resolution
$$0 \to E_{n-1} \to ... \to E_1 \to E_0 \to E \to M \to 0.$$
Conversely, by Lemma 3.1.13, $E \to M \to 0$ can be extended to a stably free resolution
$$0 \to E_{n-1} \to ... \to E_1 \to E_0 \to E \to M \to 0,$$
which, identifying $N = \ker(E \to M)$, splits as

$$0 \to E_{n-1} \to ... \to E_1 \to E_0 \to N \to 0$$
and
$$0 \to N \to E \to M \to 0.$$

In particular, we obtain a stably free resolution of N of length $n - 1$. □

The following lemma, together with the previous, will be helpful in inducting on stably free dimension.

Lemma 3.1.15. Let $0 \to M' \to M \to M'' \to 0$ be a short exact sequence of R-modules. Then, we can create a grid

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & N' \arrow[r] \arrow[d] & N \arrow[r] \arrow[d] & N'' \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & E' \arrow[r] \arrow[d] & E \arrow[r] \arrow[d] & E'' \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & M' \arrow[r] \arrow[d] & M \arrow[r] \arrow[d] & M'' \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}$$

where each row and column is exact, and E', E, E'' are free.

Proof. Throughout the proof, identify M' a a submodule of M.

Let E'' be free on a finite generating set $(\overline{m}_1, ..., \overline{m}_r)$ for M''. Choose preimages $(m_1, ..., m_r)$ of this generating set in M. Choose generators $m_{r+1}, ..., m_s$ for M'; let E' be free on those. Let $E = E' \oplus E''$.

Under this construction, the middle row is clearly exact. For commutativity, it suffices to check generators, and on generators it is immediately clear that the bottom left and bottom right squares commute. So, it remains only to contend with the top row.

Let the top row modules be the kernels of the vertical maps, so $N = \ker(E \to M)$, etc.. Since the vertical maps are surjective, we have exactness of columns.

We now set up the maps in the top row. The maps $N' \to E' \to E$ and $N \to E$ are all inclusions, so we just need to show $N' \subset N$. Indeed, any relation between the generators of M' is of course still a relation after inclusion into M. For the map $N \to N''$, we just restrict $E \to E''$. The image of N is in N'' because a relation among generators of M remains a relation among the images of those generators not sent to 0. In both cases, the square corresponds to restricting a map, so the top squares commute.

The last thing to check, exactness of the top row, follows for free by the nine lemma. It is also not bad to check directly in this case, since the middle row is free. We leave this to the interested reader.

□

Theorem 3.1.16. Let $0 \to M' \to M \to M'' \to 0$ be a short exact sequence of R-modules. If two out of three of the terms admit finite free resolutions, then the third does as well.

Proof. Create the big square as above. If anything in the top row has a finite free resolution, then so does the the term two below it. Furthermore, by Lemma 3.1, if anything in the bottom row has finite positive stably free dimension, then the term two above it has stably free dimension one less. Thus, by repeatedly passing from M's to N's, we may reduce to the case where the two modules known to have finite free resolutions have stably free dimension 0, i.e. are themselves stably free.

So, let M', M be stably free. Then, the given short exact sequence constitutes a stably free resolution of M''.

In the other two cases, M'' is stably free, and in particular projective, so $M \cong M' \oplus M''$. Direct sums of stably free modules satisfy a two-out-of-three rule, so we are done. $\square$

We are now ready to prove the main theorem of the section.

Theorem 3.1.17. If every R-module admits a finite free resolution, then every $R[x]$-module does as well.

Before we get on with the proof, we take satisfaction in using Theorem 3.1.17 to prove Theorem 3.0.1.

Proof (of Theorem 3.0.1). Every k-module is free, and in particular admits a finite free resolution. By n applications of Theorem 3.1.17, every $k[x_1, ..., x_n]$-module admits a finite free resolution. For M projective, this makes M stably free. $\square$

Proof (of Theorem 3.1.17). **Step 1: Reductions.** Let M be an $R[x]$-module. Then, M admits a finite filtration $0 = M_n \subsetneq M_{n-1} \subsetneq ... \subsetneq M_0 = M$ with subquotients $M_i/M_{i+1} \cong R[x]/\mathfrak{P}$ for $\mathfrak{P}$ prime. By Theorem 3.1.16 and induction on i, we reduce to the case that $M = R[x]/\mathfrak{P}$. In fact, by considering the exact sequence

$$0 \to \mathfrak{P} \to R[x] \to R[x]/\mathfrak{P} \to 0,$$

and noting that $R[x]$ is of course free over itself, we reduce to the case $M = \mathfrak{P}$.

The rest of this section is devoted to reducing to the case of R an integral domain, and a "minimal" one at that. Let $\mathfrak{p} = \mathfrak{P} \cap R$ be the corresponding prime downstairs. The goal is to replace R with $R/\mathfrak{p}$. Consider the exact sequence

$$0 \to \mathfrak{p}R[x] \to \mathfrak{P} \to \mathfrak{P}/(\mathfrak{p}R[x]) \to 0.$$

By hypothesis, $\mathfrak{p}$ admits a finite free resolution. Adjoining x to all the terms in that resolution yields a finite free resolution of $\mathfrak{p}R[x]$. Thus, by Theorem 3.1.16, it suffices to show that $\mathfrak{P}/(\mathfrak{p}R[x])$ admits a finite free resolution as an $R[x]$-module. We will now bootstrap with respect to the base ring of the module. Suppose $\mathfrak{P}/(\mathfrak{p}R[x])$ admits a finite free resolution

$$0 \to E_n \to ... \to E_0 \to \mathfrak{P}/(\mathfrak{p}R[x]) \to 0$$

as an $(R/\mathfrak{p})[x]$-module. The $R[x]$-module $(R/\mathfrak{p})[x]$ admits a finite free resolution, given by adjoining x to each element of a resolution of $R/\mathfrak{p}$ as an R-module, which exists by hypothesis. Since each E_i is a finite power of $(R/\mathfrak{p})[x]$, the E_i's admit finite free resolutions as well.

Let $K_n = \ker(E_n \to E_{n-1}) = \operatorname{im}(E_{n+1} \to E_n)$. Break the above resolution up into short exact sequences:

$$0 \to E_n \to E_{n-1} \to K_{n-2} \to 0$$
$$0 \to K_{n-2} \to E_{n-2} \to K_{n-3} \to 0$$
$$\ldots$$
$$0 \to K_1 \to E_1 \to K_0 \to 0$$
$$0 \to K_0 \to E_0 \to \mathfrak{P}/(\mathfrak{p}R[x]) \to 0.$$

By repeated application of Theorem 3.1.16, we conclude that $\mathfrak{P}/(\mathfrak{p}R[x])$ admits a finite free resolution as an $R[x]$-module. This completes the reduction to R an integral domain, but in fact we'll do slightly better.

Suppose towards contradiction that there is some prime of $R[x]$ not admitting a finite free resolution. By the ascending chain condition, there is some prime ideal $\mathfrak{p}$ of R which is maximal among intersections of R with prime ideals in $R[x]$ not admitting finite free resolutions; choose some $\mathfrak{P}$ above $\mathfrak{p}$ not admitting a finite free resolution. The upshot is that after quotienting by $\mathfrak{p}$, $\mathfrak{P} \cap R = \{0\}$, and any prime of $R[x]$ meeting R non-trivially admits a finite free resolution. This is the sense in which R is "minimal."

Step 2: Dirty work. Let us take inventory. We are now in the happy situation where R is a domain, and our module is in fact a prime ideal $\mathfrak{P}$ of $R[x]$ meeting R trivially, and any prime meeting R non-trivially admits a finite free resolution. A vague idea for why these reductions are progress is that primes of $R[x]$ meeting R trivially should in some sense behave a lot like primes of $k[x]$ (all of which, of course, meet k trivially), and this analogy is strengthened by the hypothesis that primes meeting R are "trivial" in that they satisfy the desired result.

Let $K = \operatorname{Frac}(R)$. The extended ideal $\mathfrak{P}K[x]$ is of course principal, say generated by some $f \in K[x]$. Clearing denominators, we can take $f \in R[x]$. The ideal $(f) \subset R[x]$ is a free $R[x]$-module, so by Theorem 3.1.16 it suffices to show $\mathfrak{P}/(f)$ admits a finite free resolution.

The module $\mathfrak{P}/(f)$ admits a filtration with subquotients $R[x]/\mathfrak{Q}$, where each $\mathfrak{Q}$ is an associated prime of $\mathfrak{P}/(f)$. As usual, Theorem 3.1.16 reduces the problem to finding finite free resolutions of the $\mathfrak{Q}$'s. Fix a generating set $g_1, ..., g_k$ for $\mathfrak{P}$. Each $g_i \in fK[x]$, so $g_i = \frac{h_i}{d_i} f$, where $h \in R[x]$ and $d \in R$. Let $d := \prod d_i$. Then, d sends each generator of $\mathfrak{P}$ into (f), so d annhiliates $\mathfrak{P}/(f)$. In particular, $d \in \mathfrak{Q}$ for each associated prime $\mathfrak{Q}$ of $\mathfrak{P}/(f)$.

Each of these associated primes meets R nontrivially, so by the minimality of R, each $\mathfrak{Q}$ admits a finite free resolution. This completes the proof. $\square$

Remark 3.1.18. Really, reduction to the situation where primes meeting R nontrivially admit finite free resolutions amounts to an induction on the Krull dimension of R. We invite the reader to ponder whether a proof that proceeds explicitly by induction on Krull dimension of R would be easier or harder to understand.

3.2 UCEP implies stably free is free

Definition 3.2.1. Let R be a ring, and let $\mathbf{x}$ be a vector in R^n. We say $\mathbf{x}$ is *unimodular* if its entries generate the unit ideal.

Definition 3.2.2. We say a vector $\mathbf{x}$ has the *extension property* if $\mathbf{x}$ may be realized as the first column of an invertible matrix $M \in \mathrm{GL}_n(R)$.

The determinant of a matrix is contained in the ideal generated by its first column entries. Thus, a necessary condition to have the extension property is to be unimodular. We shall soon see that it is very profitable to ask when the converse holds.

Definition 3.2.3. We say a ring R has the *unimodular column extension property* if every unimodular vector in R has the extension property.

Theorem 3.2.4 (3.0.3)**.** Suppose A has the unimodular column extension property. Then, stably free modules over A are free.

Proof. With the unimodular column extension property in hand, we may proceed essentially the same way as one would in the context of vector spaces. Let E be a stably free module, so $E \oplus A^m \cong A^n$. By induction on m, it suffices to consider the case of $m = 1$, so $E \oplus A \cong A^n$. Let $\varphi : A^n \to E \oplus A$ be such an isomorphism. Let $u_1 = \varphi^{-1}(0, 1)$. Restricting to the second factor, we get a linear map realizing 1 as a linear combination of the entries of u_1; we conclude that u is unimodular. By the unimodular column extension property, we may complete u_1 to a basis $u_1, ..., u_n$ for A^n. Let $\varphi(u_i) = (\varepsilon_i, a_i)$. For $i > 1$, define $u_i' := u_i - a_i u_1$. The set $\{u_1, u_2', ..., u_n'\}$ still forms a basis for A^n, now with the added property that $\varphi(u_i')|_A = 0$. Restricting φ to $\langle u_2', ..., u_n' \rangle$, we obtain an isomorphism $A^{n-1} \xrightarrow{\sim} E$. □

3.3 Polynomial rings have the UCEP

Seeing that the unimodular column extension property is just the thing needed to trivialize stably free modules, we now spend considerable effort establishing when it holds. We warm up with the case of Euclidean domains.

Lemma 3.3.1. Euclidean domains satisfy the unimodular column extension property.

Proof. Let R be a Euclidean domain, with size function σ. Let $f_1, ..., f_n$ be the entries of a unimodular vector. We proceed by induction on $\max\{\sigma(f_i)\}$, with the number of f_i achieving the maximum as a tiebreaker. As a base case, when the maximum is 1, and this maximum is only achieved in one entry, then the vector is all zeros except for one unit. Such vectors clearly have the extension property.

So, assume $\max\{\sigma(f_i)\} > 1$. Let f_i be an entry achieving the maximum. By unimodularity, another entry is non-zero, say f_j. If $\max\{\sigma(f_i)\} = 1$ and there are multiple achievers, then we may similarly take a maximizer and another non-zero entry. Divide f_i by f_j with remainder; replace f_1 by the remainder r using the corresponding row operation. The extension property is invariant under row operations, so the proof is complete. □

As discussed in Remark 3.1.3, the unimodular column extension property is not preserved by adjoining a variable, so a general induction on the number of variables would not work. However, the proof still proceeds by induction; it will just use as an inductive hypothesis that R is a polynomial ring. We now build up the lemmas that will go into the proof. We start in the local setting.

Lemma 3.3.2. Let $(R, \mathfrak{m})$ be a local ring, and let $\mathbf{f}$ be a unimodular vector with entries in $R[x]$. Suppose $\mathbf{f}$ has a monic entry. Then, $\mathbf{f}$ has the extension property.

Proof. At some point in the main proof, we will want to use three separate entries of $\mathbf{f}$. So, we briefly dispense with $n = 1, 2$ first. For $n = 1$, the 1 by 1 vector is already a 1 by 1 matrix. For $n = 2$, suppose that $\langle f_1, f_2 \rangle$ is unimodular. Then, there is some sum $g_1 f_1 + g_2 f_2 = 1$. The matrix

$$\begin{bmatrix} f_1 & g_2 \\ f_2 & -g_1 \end{bmatrix}$$

suffices.

We now assume $n \geq 3$. The goal is to reduce $\mathbf{f}$ to a standard basis vector using row operations. Formally, we will proceed by induction on the minimal degree of any monic entry. Let $f_1, ..., f_n$ denote the entries of $\mathbf{f}$, and, without loss of generality, let f_1 be monic of degree d. If $d = 0$, then f_1 is a unit, and so $\mathbf{f}$ has the extension property. Now let $d > 0$. Since f_1 is monic, we may divide by it with remainder, and thus we assume that all other entries have strictly lower degree.

To complete the induction, we want to show that there is some linear combination $\sum g_i f_i$, with one of the g_i equal to 1, yielding a monic polynomial of degree at most $d - 1$. Fix a bad index i, which will be where $g_i = 1$. Consider the set of possible leading coefficients of polynomials of degree $d - 1$ obtained as linear combinations of the entries away from i. By adding and scaling the vectors $(g_1, ..., \hat{g}_i, ..., g_n)$, one sees that the set of such coefficients forms an ideal of R; call the ideal I. We now try to show that I is the unit ideal.

We claim that I contains the coefficients of the f_j for $j \neq 1, i$. Write

$$\begin{aligned} f_j(x) &= \phantom{x^d+{}} c_{d-1}x^{d-1} + ... + c_1 x + c_0 \\ f_1(x) &= x^d + b_{d-1}x^{d-1} + ... + b_1 x + b_0. \end{aligned}$$

Containment $c_{d-1} \in I$ is obvious. Now assume all c_ℓ for $\ell > k$ are in I. Divide $x^{d-1-k} f_j(x)$ by $f_1(x)$ with remainder. At each step in the division, the leading coefficient is some linear combination of c_ℓ for $\ell > k$. The x^{d-1} coefficient starts as c_k and at each step the x^{d-1}-coefficient is the product of some b_ℓ and the leading coefficient. At the end of the division, the remainder polynomial will on one hand have been expressed as a linear combination of f_1 and f_j, and on the other hand will have leading coefficient of the form $c_k + C$, where $C \in I$. Thus, $c_k \in I$.

By unimodularity, there is a linear combination $\sum g_i f_i = 1$. Upon passing to $(R/\mathfrak{m})[x]$, the relation still holds, and f_1 is not a unit, so there is some other non-zero entry. Returning to $R[x]$, we conclude that some other entry has a unit coefficient. Without loss of generality, let f_2 be such an entry, and take bad index $i = 3$. We conclude $I = R$, so there is linear combination of f_j, $j \neq i$ monic of degree $d - 1$. Let f_i have leading coefficient a. We may perform the row operation of subtracting $a - 1$ times the monic linear combination to yield a monic entry with degree $d - 1$, completing the induction. $\square$

Successive row operations can be encoded as products of elementary matrices, and under this encoding the above proof may be reinterpreted as the construction of an invertible matrix M such that $M\mathbf{f} = e_1$.

Definition 3.3.3. In general, we will call two vectors $\mathbf{x}$ and $\mathbf{y}$ *equivalent* and write $\mathbf{x} \sim \mathbf{y}$ if they differ by multiplication by an invertible matrix.

Equivalently, we say vectors are equivalent if they are in the same orbit of the $\mathrm{GL}_n(R)$-action on R^n.

Example 3.3.4. Let $\mathbf{f}(x)$ denote a vector with entries f_i in $R[x]^n$, and let $\mathbf{f}(0)$ denote corresponding vector in R^n obtained by coordinate-wise evaluation. In the case where R is local and $\mathbf{f}$ has a monic entry, we now know $\mathbf{f} \sim e_1$. Over a local ring, every unimodular vector has a unit as an entry, and so by row operations is equivalent to e_1. By transitivity, we have $\mathbf{f}(x) \sim \mathbf{f}(0)$, and in fact the 0 can be replaced by any element of R.

Our next goal is to show that the local hypothesis on R can be removed.

Lemma 3.3.5. Let R be a domain and let $\mathfrak{p}$ be a prime of R. Let $\mathbf{f}$ be a vector in $R[x]$. If $\mathbf{f}(x) \sim \mathbf{f}(0)$ over the localization $R_{\mathfrak{p}}[x]$, then there is a $c \in R \backslash \mathfrak{p}$ for which $\mathbf{f}(cx + y) \sim \mathbf{f}(y)$ over $R[x, y]$.

Remark 3.3.6. The inclusion of the additional variable y in this lemma statement may seem strange; it will provide important maneuvering space in the proof of Lemma 3.3.7.

Proof. Since $\mathbf{f}(x) \sim \mathbf{f}(0)$, we get by two changes of coordinates that $\mathbf{f}(x + y) \sim \mathbf{f}(0) \sim \mathbf{f}(y)$, and so by transitivity $\mathbf{f}(x + y) \sim \mathbf{f}(y)$. Let M be a matrix realizing this equivalence.

Evaluating the equation $M(x, y)\mathbf{f}(x+y) = \mathbf{f}(y)$ at $x = 0$ yields $M(0, y)\mathbf{f}(y) \sim \mathbf{f}(y)$. Let $N(x, y) = M(0, y)^{-1}M(x, y)$. Then, N also realizes an equivalence between $\mathbf{f}(x + y)$ and $\mathbf{f}(y)$, and furthermore $N(0, y) = I_n$. Write

$$N = I_n + xH(x, y).$$

All the matrices involved still have coefficients in $R_{\mathfrak{p}}[x, y]$; we want something with coefficients in $R[x, y]$. Let $d \in R \backslash \mathfrak{p}$ be a denominator for H, i.e. a number such that dH has entries in $R[x, y]$. Then, $N(dx, y) = I_n + dxH(dx, y)$ has entries in $R[x, y]$, and $N(dx, y)\mathbf{f}(dx + y) = \mathbf{f}(y)$, as desired. $\square$

We now apply the above lemma to obtain a global result.

Lemma 3.3.7. Let R be a domain, and let $\mathbf{f} \in R[x]^n$ be a unimodular vector with a monic entry. Then, $\mathbf{f}(x) \sim \mathbf{f}(0)$.

Proof. Consider the set J of c such that $\mathbf{f}(cx+y) \sim \mathbf{f}(y)$. We claim J is an ideal. Indeed, given $r \in R$ and $c \in J$, we may take the equation $M(x, y)\mathbf{f}(cx+y) = \mathbf{f}(y)$ and just substitute $(x, y) \mapsto (rx, y)$. Given $c_1, c_2 \in J$, we have $\mathbf{f}(c_1x + y) \sim \mathbf{f}(y)$ by hypothesis, and applying the change of coordinates $(x, y) \mapsto (x, y + c_1x)$ to the other equivalence we get $\mathbf{f}((c_1 + c_2)x + y) \sim \mathbf{f}(c_1x + y)$. Transitivity gives $\mathbf{f}((c_1 + c_2)x + y) \sim \mathbf{f}(y)$, completing the proof that J is an ideal.

It is worth noting that the change of coordinates used to prove that J is closed under sums would not have been possible if we had instead considered the set of c such that $\mathbf{f}(cx) \sim \mathbf{f}(0)$. The above proof of sum-closure of J is the reason, foreshadowed in Remark 3.3.6, that we introduced the variable y in the statement of Lemma 3.3.5.

By Lemma 3.3.2, $\mathbf{f}(x) \sim \mathbf{f}(0)$ over the localization $R_\mathfrak{p}$ at each prime $\mathfrak{p}$ of R. Thus, by Lemma 3.3.5, for each $\mathfrak{p}$ there is some $c_\mathfrak{p} \in R \backslash \mathfrak{p}$ for which $\mathbf{f}(cx + y) \sim \mathbf{f}(y)$. Thus, J is not contained in any prime ideal $\mathfrak{p}$, and so J is the unit ideal. In particular, $\mathbf{f}(x + y) \sim \mathbf{f}(y)$. We may at last substitute $y = 0$ to yield the desired $\mathbf{f}(x) \sim \mathbf{f}(0)$. $\square$

Theorem 3.3.8. Let k be a field. Then, $k[x_1, ..., x_n]$ has the unimodular column extension property.

Proof. The proof proceeds by induction on the number of variables. For zero variables, this is just linear algebra. In the case of r variables, write $R = k[x_1, ..., x_{r-1}]$ and correspondingly write $R[x] = k[x_1, ..., x_r]$. Let $\mathbf{f}$ be a vector in $R[x]^n$. We would like to massage $\mathbf{f}$ to have a monic entry (as a univariate polynomial in x_r), so that we may apply Lemma 3.3.7. We use the "Noether normalization trick," performing the change of variables

$$\begin{aligned} &y_r = x_r \\ &y_i = x_i - x_r^m, \qquad x_i = y_i + y_r^m, \end{aligned}$$

where m is sufficiently large (twice the maximum of the degrees of the total degrees of the entries should suffice). Every entry of the resulting polynomial in the y_i's is either constant in y_r or has monic leading y_r coefficient. If every entry is constant in y_r, then $\mathbf{f}$ has the extension property by the inductive hypothesis. Otherwise, we have created our desired monic entry, allowing us to apply Lemma 3.3.7 and complete the proof. $\square$

Corollary 3.3.9 (Quillen-Suslin theorem)**.** Finitely generated projective modules over $k[x_1, ..., x_n]$ are free.

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